

Amendment No. 2
To
AIS-049(Rev.1):2016 Electric Power Train Vehicles - CMVR Type Approval
for Electric
Power Train Vehicles

1.0 Clause No 3.4

Substitute following text for existing text:

3.4 Coast Down test

Coast down tests for coefficient of aerodynamic and rolling resistance shall be done as per AIS 137 standard as amended from time to time for L, M1, M2 (up to 3500 kg GVW) & N1 Category Electric Power Train Vehicles till the time revision of AIS 039 Rev 1:2015 and AIS 040 (Rev 1):2015 is notified.

For M2(above 3500 kg GVW) and M3, N2 and N3 Category Electric Power Train Vehicles, Coast down tests for coefficient of aerodynamic and rolling resistance shall be done as per IS 14785 as amended time to time till the time revision of AIS 039 Rev 1:2015 and AIS 040 (Rev 1):2015 is notified.

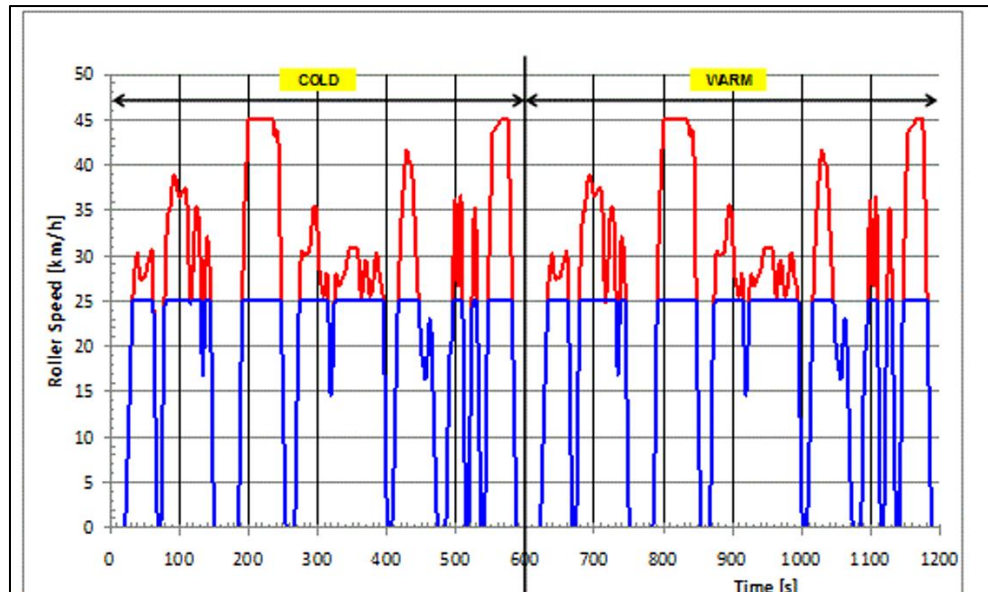
2.0 Clause No C1.58 (iv)

Substitute following text for existing text:

(iv) For M2 (with GVW above 3500 kg) and M3 vehicles: unladen mass increased by 75 kg x 50% of Seating Capacity. For school bus approval, unladen mass increased by 50 kg x 50% of Seating Capacity shall be considered.

Add new Annex: F after annexure E as below and rename existing Annex F and G to G and H respectively

Driving cycle WMTC RST 25 and RST 45



Note: The blue line depicts RST25, i.e. the truncated vehicle speed trace limited to 25 km/h, applicable for vehicles with a maximum design vehicle speed of 25 km/h.

The blue line, extended by the red line for speeds over 25 km/h, depicts RST 45, i.e. the truncated vehicle speed trace limited to 45 km/h, applicable for vehicles with a maximum design vehicle speed of 45 km/h.

The desired vehicle speed trace WMTC shown in above Figure is applicable for vehicles with a maximum design vehicle speed (if applicable at 25 km/h, at 45 km/h or 50 km/h)) and consists of the desired vehicle speed trace WMTC stage 1. part 1 for L1 category (speed less than 50 km/h) vehicles driven once cold followed by the same desired vehicle speed trace driven with a warm propulsion unit.

The WMTC for vehicles with a low maximum design vehicle speed or maximum net or continuous rated power lasts 1200 seconds and consists of two equivalent parts to be carried out without interruption.

The characteristic driving conditions (idling, acceleration, steady vehicle speed, deceleration, etc.) of the WMTC for vehicles with a maximum design vehicle speed (if applicable at 25 km/h, at 45 km/h, or at 50 km/h) are set out in the following tables

RST 25

WMTC. part 1. applicable for vehicles with a maximum design vehicle speed (25 km/h), cold or warm, 0 to 180 s

time in s	roller speed in km/h	phase indicators				time in s	roller speed in km/h	phase indicators				time in s	roller speed in km/h	phase indicators			
		stop	acc	cruise	dec			stop	acc	cruise	dec			stop	acc	cruise	dec
0	0	X				33	25					66	9.3				X
1	0	X				34	25					67	4.8				X
2	0	X				35	25					68	1.9				X
3	0	X				36	25					69	0	X			
4	0	X				37	25					70	0	X			
5	0	X				38	25					71	0	X			
6	0	X				39	25			X		72	0	X			
7	0	X				40	25			X		73	0	X			
8	0	X				41	25			X		74	1.7		X		
9	0	X				42	25			X		75	5.8		X		
10	0	X				43	25			X		76	11.8		X		
11	0	X				44	25			X		77	17.3		X		
12	0	X				45	25			X		78	22		X		
13	0	X				46	25			X		79	25				
14	0	X				47	25			X		80	25				
15	0	X				48	25			X		81	25				
16	0	X				49	25			X		82	25				
17	0	X				50	25			X		83	25				
18	0	X				51	25			X		84	25				
19	0	X				52	25			X		85	25				
20	0	X				53	25			X		86	25				
21	0	X				54	25			X		87	25				
22	1		X			55	25			X		88	25				
23	2.6		X			56	25			X		89	25				
24	4.8		X			57	25			X		90	25				
25	7.2		X			58	25			X		91	25			X	
26	9.6		X			59	25			X		92	25			X	
27	12		X			60	25				X	93	25			X	
28	14.3		X			61	25					94	25			X	
29	16.6		X			62	25					95	25			X	
30	18.9		X			63	23				X	96	25			X	
31	21.2		X			64	18.6				X	97	25			X	
32	23.5		X			65	14.1				X	98	25			X	

time in s	roller speed in km/h	phase indicators				time in s	roller speed in km/h	phase indicators				time in s	roller speed in km/h	phase indicators			
		stop	acc	cruise	dec			stop	acc	cruise	dec			stop	acc	cruise	dec
99	25			X		126	25					154	0	X			
100	25			X		127	25					155	0	X			
101	25			X		128	25					156	0	X			
102	25			X		129	25					157	0	X			
103	25			X		130	25					158	0	X			
104	25			X		131	25					159	0	X			
105	25			X		132	22.1				X	160	0	X			
106	25			X		133	18.6				X	161	0	X			
107	25			X		134	16.8		X			162	0	X			
108	25			X		135	17.7		X			163	0	X			
109	25			X		136	21.1		X			164	0	X			
110	25					137	25					165	0	X			
111	25					138	25					166	0	X			
112	25					139	25					167	0	X			
113	25					140	25					168	0	X			
114	25					141	25					169	0	X			
115	25					142	25					170	0	X			
116	24.7			X		143	25					171	0	X			
117	25			X		144	25					172	0	X			
118	25			X		145	25					173	0	X			
119	25			X		146	20.3				X	174	0	X			
120	25			X		147	15				X	175	0	X			
						148	9.7				X	176	0	X			
121	25			X		149	5				X	177	0	X			
122	25			X		150	1.6				X	178	0	X			
123	25			X		151	0	X				179	0	X			
124	25			X		152	0	X				180	0	X			
125	25					153	0	X									

WMTC. part 1. applicable for vehicles with a maximum design vehicle speed (25 km/h), cold or warm, 181 to 360 s

time in s	roller speed in km/h	phase indicators				time in s	roller speed in km/h	phase indicators				time in s	roller speed in km/h	phase indicators			
		stop	acc	cruise	dec			stop	acc	cruise	dec			stop	acc	cruise	dec
181	0	X				211	25			X		241	25			X	
182	0	X				212	25			X		242	25				
183	0	X				213	25			X		243	25				
184	0	X				214	25			X		244	25				
185	0.4		X			215	25			X		245	25				
186	1.8		X			216	25			X		246	25				
187	5.4		X			217	25			X		247	25				
188	11.1		X			218	25			X		248	21.8				X
189	16.7		X			219	25			X		249	17.2				X
190	21.3		X			220	25			X		250	13.7				X
191	24.8		X			221	25			X		251	10.3				X
192	25					222	25			X		252	7				X
193	25					223	25			X		253	3.5				X
194	25					224	25			X		254	0	X			
195	25					225	25			X		255	0	X			
196	25					226	25			X		256	0	X			
197	25					227	25			X		257	0	X			
198	25					228	25			X		258	0	X			
199	25					229	25			X		259	0	X			
200	25					230	25			X		260	0	X			
201	25					231	25			X		261	0	X			
202	25					232	25			X		262	0	X			
203	25			X		233	25			X		263	0	X			
204	25			X		234	25			X		264	0	X			
205	25			X		235	25			X		265	0	X			
206	25			X		236	25			X		266	0	X			
207	25			X		237	25			X		267	0.5		X		
208	25			X		238	25			X		268	2.9		X		
209	25			X		239	25			X		269	8.2		X		
210	25			X		240	25			X		270	13.2		X		

time in s	roller speed in km/h	phase indicators				time in s	roller speed in km/h	phase indicators				time in s	roller speed in km/h	phase indicators			
		stop	acc	cruise	dec			stop	acc	cruise	dec			stop	acc	cruise	dec
271	17.8		X			301	25			X		331	25			X	
272	21.4		X			302	25			X		332	25			X	
273	24.1		X			303	25			X		333	25			X	
274	25					304	25			X		334	25			X	
275	25					305	25			X		335	25			X	
276	25					306	25			X		336	25			X	
277	25			X		307	25			X		337	25			X	
278	25			X		308	25			X		338	25			X	
279	25			X		309	25			X		339	25			X	
280	25			X		310	25			X		340	25			X	
281	25			X		311	25			X		341	25			X	
282	25			X		312	25			X		342	25			X	
283	25			X		313	25			X		343	25			X	
284	25			X		314	25					344	25			X	
285	25			X		315	25					345	25			X	
286	25			X		316	22.7				X	346	25			X	
287	25			X		317	19				X	347	25			X	
288	25			X		318	16				X	348	25			X	
289	25			X		319	14.6		X			349	25			X	
290	25			X		320	15.2		X			350	25			X	
291	25			X		321	16.9		X			351	25			X	
292	25			X		322	19.3		X			352	25			X	
293	25			X		323	22		X			353	25			X	
294	25			X		324	24.6		X			354	25			X	
295	25			X		325	25					355	25			X	
296	25			X		326	25					356	25			X	
297	25			X		327	25			X		357	25			X	
298	25			X		328	25			X		358	25			X	
299	25			X		329	25			X		359	25			X	
300	25			X		330	25			X		360	25			X	

WMTC. part 1. applicable for vehicles with a maximum design vehicle speed (25 km/h), cold or warm, 361 to 540 s

time in s	roller speed in km/h	phase indicators				time in s	roller speed in km/h	phase indicators				time in s	roller speed in km/h	phase indicators			
		stop	acc	cruise	dec			stop	acc	cruise	dec			stop	acc	cruise	dec
361	25			X		391	25			X		421	25		X		
362	25			X		392	25					422	25		X		
363	25			X		393	25					423	25		X		
364	25			X		394	25					424	25		X		
365	25			X		395	24.9				X	425	25		X		
366	25			X		396	21.4				X	426	25		X		
367	25			X		397	15.9				X	427	25		X		
368	25			X		398	9.9				X	428	25		X		
369	25			X		399	4.9				X	429	25			X	
370	25			X		400	2.1				X	430	25			X	
371	25			X		401	0.9				X	431	25			X	
372	25			X		402	0	X				432	25			X	
373	25			X		403	0	X				433	25			X	
374	25			X		404	0	X				434	25			X	
375	25			X		405	0	X				435	25			X	
376	25			X		406	0	X				436	25				
377	25			X		407	0	X				437	25				
378	25			X		408	1.2		X			438	25				
379	25			X		409	3.2		X			439	25				
380	25			X		410	5.9		X			440	25				
381	25			X		411	8.8		X			441	25				
382	25			X		412	12		X			442	25				
383	25			X		413	15.4		X			443	25				
384	25			X		414	18.9		X			444	25				
385	25			X		415	22.1		X			445	25				
386	25			X		416	24.7		X			446	25				
387	25			X		417	25					447	23.4				X
388	25			X		418	25					448	21.8				X
389	25			X		419	25					449	20.3				X
390	25			X		420	25					450	19.3				X

time in s	roller speed in km/h	phase indicators				time in s	roller speed in km/h	phase indicators				time in s	roller speed in km/h	phase indicators			
		stop	acc	cruise	dec			stop	acc	cruise	dec			stop	acc	cruise	dec
451	18.7				X	481	0	X				511	16.7				X
452	18.3				X	482	0	X				512	10.7				X
453	17.8				X	483	0	X				513	4.7				X
454	17.4				X	484	0	X				514	1.2				X
455	16.8				X	485	0	X				515	0	X			
456	16.3			X		486	1.4		X			516	0	X			
457	16.5			X		487	4.5		X			517	0	X			
458	17.6			X		488	8.8		X			518	0	X			
459	19.2			X		489	13.4		X			519	3		X		
460	20.8			X		490	17.3		X			520	8.2		X		
461	22.2			X		491	19.2		X			521	14.3		X		
462	23			X		492	19.7		X			522	19.3		X		
463	23				X	493	19.8		X			523	23.5		X		
464	22				X	494	20.7		X			524	25				
465	20.1				X	495	23.7		X			525	25				
466	17.7				X	496	25					526	25				
467	15				X	497	25					527	25				
468	12.1				X	498	25					528	25				
469	9.1				X	499	25					529	25				
470	6.2				X	500	25					530	25				
471	3.6				X	501	25					531	23.2				X
472	1.8				X	502	25					532	18.5				X
473	0.8				X	503	25					533	13.8				X
474	0	X				504	25					534	9.1				X
475	0	X				505	25					535	4.5				X
476	0	X				506	25					536	2.3				X
477	0	X				507	25					537	0	X			
478	0	X				508	25					538	0	X			
479	0	X				509	25					539	0	X			
480	0	X				510	23.1				X	540	0				

WMTC. part 1.applicable for vehicles with a maximum design vehicle speed (25 km/h), cold or warm, 541 to 600 s

<i>time in s</i>	<i>roller speed in km/h</i>	<i>phase indicators</i>			
		<i>stop</i>	<i>acc</i>	<i>cruise</i>	<i>dec</i>
541	0	X			
542	2.8		X		
543	8.1		X		
544	14.3		X		
545	19.2		X		
546	23.5		X		
547	25				
548	25				
549	25				
550	25				
551	25				
552	25				
553	25			X	
554	25			X	
555	25			X	
556	25			X	
557	25			X	
558	25			X	
559	25			X	
560	25			X	
561	25			X	
562	25			X	
563	25			X	
564	25			X	
565	25			X	
566	25			X	
567	25			X	
568	25			X	
569	25			X	
570	25			X	
571	25			X	
572	25			X	
573	25				

<i>time in s</i>	<i>roller speed in km/h</i>	<i>phase indicators</i>			
		<i>stop</i>	<i>acc</i>	<i>cruise</i>	<i>dec</i>
574	25				
575	25				
576	25				
577	25				
578	25				
579	25				
580	25				
581	25				
582	21.8				X
583	17.7				X
584	13.5				X
585	9.4				X
586	5.6				X
587	2.1				X
588	0	X			
589	0	X			
590	0	X			
591	0	X			
592	0	X			
593	0	X			
594	0	X			
595	0	X			
596	0	X			
597	0	X			
598	0	X			
599	0	X			
600	0	X			

RST 45

WMTC. part 1. applicable for vehicles with a maximum design vehicle speed (where applicable truncated at 45km/h & 50km/h respectively) ,cold or warm, 0 to 180 s

time in s	roller speed in km/h	phase indicators				time in s	roller speed in km/h	phase indicators				time in s	roller speed in km/h	phase indicators			
		stop	acc	cruise	dec			stop	acc	cruise	dec			stop	acc	cruise	dec
0	0	X				33	25.6		X			66	9.3				X
1	0	X				34	27.1		X			67	4.8				X
2	0	X				35	28		X			68	1.9				X
3	0	X				36	28.7		X			69	0	X			
4	0	X				37	29.2		X			70	0	X			
5	0	X				38	29.8		X			71	0	X			
6	0	X				39	30.3			X		72	0	X			
7	0	X				40	29.6			X		73	0	X			
8	0	X				41	28.7			X		74	1.7		X		
9	0	X				42	27.9			X		75	5.8		X		
10	0	X				43	27.4			X		76	11.8		X		
11	0	X				44	27.3			X		77	17.3		X		
12	0	X				45	27.3			X		78	22		X		
13	0	X				46	27.4			X		79	26.2		X		
14	0	X				47	27.5			X		80	29.4		X		
15	0	X				48	27.6			X		81	31.1		X		
16	0	X				49	27.6			X		82	32.9		X		
17	0	X				50	27.6			X		83	34.7		X		
18	0	X				51	27.8			X		84	34.8		X		
19	0	X				52	28.1			X		85	34.8		X		
20	0	X				53	28.5			X		86	34.9		X		
21	0	X				54	28.9			X		87	35.4		X		
22	1		X			55	29.2			X		88	36.2		X		
23	2.6		X			56	29.4			X		89	37.1		X		
24	4.8		X			57	29.7			X		90	38		X		
25	7.2		X			58	30			X		91	38.7			X	
26	9.6		X			59	30.5			X		92	38.9			X	
27	12		X			60	30.6				X	93	38.9			X	
28	14.3		X			61	29.6				X	94	38.8			X	
29	16.6		X			62	26.9				X	95	38.5			X	
30	18.9		X			63	23				X	96	38.1			X	
31	21.2		X			64	18.6				X	97	37.5			X	
32	23.5		X			65	14.1				X	98	37			X	

time in s	roller speed in km/h	phase indicators				time in s	roller speed in km/h	phase indicators				time in s	roller speed in km/h	phase indicators			
		stop	acc	cruise	dec			stop	acc	cruise	dec			stop	acc	cruise	dec
99	36.7			X		126	35.2				X	154	0	X			
100	36.5			X		127	34.7				X	155	0	X			
101	36.5			X		128	33.9				X	156	0	X			
102	36.6			X		129	32.4				X	157	0	X			
103	36.8			X		130	29.8				X	158	0	X			
104	37			X		131	26.1				X	159	0	X			
105	37.1			X		132	22.1				X	160	0	X			
106	37.3			X		133	18.6				X	161	0	X			
107	37.4			X		134	16.8		X			162	0	X			
108	37.5			X		135	17.7		X			163	0	X			
109	37.4			X		136	21.1		X			164	0	X			
110	36.9				X	137	25.4		X			165	0	X			
111	36				X	138	29.2		X			166	0	X			
112	34.8				X	139	31.6		X			167	0	X			
113	31.9				X	140	32.1				X	168	0	X			
114	29				X	141	31.6				X	169	0	X			
115	26.9				X	142	30.7				X	170	0	X			
116	24.7			X		143	29.7				X	171	0	X			
117	25.4			X		144	28.1				X	172	0	X			
118	26.4			X		145	25				X	173	0	X			
119	27.7			X		146	20.3				X	174	0	X			
120	29.4			X		147	15				X	175	0	X			
						148	9.7				X	176	0	X			
121	31.2			X		149	5				X	177	0	X			
122	33			X		150	1.6				X	178	0	X			
123	34.4			X		151	0	X				179	0	X			
124	35.2			X		152	0	X				180	0	X			
125	35.4				X	153	0	X									

WMTC. part 1. applicable for vehicles with a maximum design vehicle speed (where applicable truncated at 45 km/h & 50km/h. respectively), cold or warm, 181 to 360 s

time in s	roller speed in km/h	phase indicators				time in s	roller speed in km/h	phase indicators				time in s	roller speed in km/h	phase indicators			
		stop	acc	cruise	dec			stop	acc	cruise	dec			stop	acc	cruise	dec
181	0	X				211	45			X		241	43.9			X	
182	0	X				212	45			X		242	43.8				X
183	0	X				213	45			X		243	43				X
184	0	X				214	45			X		244	40.9				X
185	0.4		X			215	45			X		245	36.9				X
186	1.8		X			216	45			X		246	32.1				X
187	5.4		X			217	45			X		247	26.6				X
188	11.1		X			218	45			X		248	21.8				X
189	16.7		X			219	45			X		249	17.2				X
190	21.3		X			220	45			X		250	13.7				X
191	24.8		X			221	45			X		251	10.3				X
192	28.4		X			222	45			X		252	7				X
193	31.8		X			223	45			X		253	3.5				X
194	34.6		X			224	45			X		254	0	X			
195	36.3		X			225	45			X		255	0	X			
196	37.8		X			226	45			X		256	0	X			
197	39.6		X			227	45			X		257	0	X			
198	41.3		X			228	45			X		258	0	X			
199	43.3		X			229	45			X		259	0	X			
200	45					230	45			X		260	0	X			
201	45					231	45			X		261	0	X			
202	45					232	45			X		262	0	X			
203	45			X		233	45			X		263	0	X			
204	45			X		234	45			X		264	0	X			
205	45			X		235	45			X		265	0	X			
206	45			X		236	44.4			X		266	0	X			
207	45			X		237	43.5			X		267	0.5		X		
208	45			X		238	43.2			X		268	2.9		X		
209	45			X		239	43.3			X		269	8.2		X		
210	45			X		240	43.7			X		270	13.2		X		

time in s	roller speed in km/h	phase indicators				time in s	roller speed in km/h	phase indicators				time in s	roller speed in km/h	phase indicators			
		stop	acc	cruise	dec			stop	acc	cruise	dec			stop	acc	cruise	dec
271	17.8		X			301	30.6			X		331	26.6			X	
272	21.4		X			302	29			X		332	26.8			X	
273	24.1		X			303	27.8			X		333	27			X	
274	26.4		X			304	27.2			X		334	27.2			X	
275	28.4		X			305	26.9			X		335	27.4			X	
276	29.9		X			306	26.5			X		336	27.5			X	
277	30.5			X		307	26.1			X		337	27.7			X	
278	30.5			X		308	25.7			X		338	27.9			X	
279	30.3			X		309	25.5			X		339	28.1			X	
280	30.2			X		310	25.7			X		340	28.3			X	
281	30.1			X		311	26.4			X		341	28.6			X	
282	30.1			X		312	27.3			X		342	29.1			X	
283	30.1			X		313	28.1			X		343	29.6			X	
284	30.2			X		314	27.9				X	344	30.1			X	
285	30.2			X		315	26				X	345	30.6			X	
286	30.2			X		316	22.7				X	346	30.8			X	
287	30.2			X		317	19				X	347	30.8			X	
288	30.5			X		318	16				X	348	30.8			X	
289	31			X		319	14.6		X			349	30.8			X	
290	31.9			X		320	15.2		X			350	30.8			X	
291	32.8			X		321	16.9		X			351	30.8			X	
292	33.7			X		322	19.3		X			352	30.8			X	
293	34.5			X		323	22		X			353	30.8			X	
294	35.1			X		324	24.6		X			354	30.9			X	
295	35.5			X		325	26.8		X			355	30.9			X	
296	35.6			X		326	27.9		X			356	30.9			X	
297	35.4			X		327	28			X		357	30.8			X	
298	35			X		328	27.7			X		358	30.4			X	
299	34			X		329	27.1			X		359	29.6			X	
300	32.4			X		330	26.8			X		360	28.4			X	

WMTC. part 1. applicable for vehicles with a maximum design vehicle speed (where applicable truncated at 45 km/h & 50km/h. respectively), cold or warm, 361 to 540 s

time in s	roller speed in km/h	phase indicators				time in s	roller speed in km/h	phase indicators				time in s	roller speed in km/h	phase indicators			
		stop	acc	cruise	dec			stop	acc	cruise	dec			stop	acc	cruise	dec
361	27.1			X		391	27.2			X		421	34		X		
362	26			X		392	26.9				X	422	35.4		X		
363	25.4			X		393	26.4				X	423	36.5		X		
364	25.5			X		394	25.7				X	424	37.5		X		
365	26.3			X		395	24.9				X	425	38.6		X		
366	27.3			X		396	21.4				X	426	39.6		X		
367	28.3			X		397	15.9				X	427	40.7		X		
368	29.2			X		398	9.9				X	428	41.4		X		
369	29.5			X		399	4.9				X	429	41.7			X	
370	29.4			X		400	2.1				X	430	41.4			X	
371	28.9			X		401	0.9				X	431	40.9			X	
372	28.1			X		402	0	X				432	40.5			X	
373	27.1			X		403	0	X				433	40.2			X	
374	26.3			X		404	0	X				434	40.1			X	
375	25.7			X		405	0	X				435	40.1			X	
376	25.5			X		406	0	X				436	39.8				X
377	25.6			X		407	0	X				437	38.9				X
378	25.9			X		408	1.2		X			438	37.4				X
379	26.3			X		409	3.2		X			439	35.8				X
380	26.9			X		410	5.9		X			440	34.1				X
381	27.6			X		411	8.8		X			441	32.5				X
382	28.4			X		412	12		X			442	30.9				X
383	29.3			X		413	15.4		X			443	29.4				X
384	30.1			X		414	18.9		X			444	27.9				X
385	30.4			X		415	22.1		X			445	26.5				X
386	30.2			X		416	24.7		X			446	25				X
387	29.5			X		417	26.8		X			447	23.4				X
388	28.6			X		418	28.7		X			448	21.8				X
389	27.9			X		419	30.6		X			449	20.3				X
390	27.5			X		420	32.4		X			450	19.3				X

time in s	roller speed in km/h	phase indicators				time in s	roller speed in km/h	phase indicators				time in s	roller speed in km/h	phase indicators			
		stop	acc	cruise	dec			stop	acc	cruise	dec			stop	acc	cruise	dec
451	18.7				X	481	0	X				511	16.7				X
452	18.3				X	482	0	X				512	10.7				X
453	17.8				X	483	0	X				513	4.7				X
454	17.4				X	484	0	X				514	1.2				X
455	16.8				X	485	0	X				515	0	X			
456	16.3			X		486	1.4		X			516	0	X			
457	16.5			X		487	4.5		X			517	0	X			
458	17.6			X		488	8.8		X			518	0	X			
459	19.2			X		489	13.4		X			519	3		X		
460	20.8			X		490	17.3		X			520	8.2		X		
461	22.2			X		491	19.2		X			521	14.3		X		
462	23			X		492	19.7		X			522	19.3		X		
463	23				X	493	19.8		X			523	23.5		X		
464	22				X	494	20.7		X			524	27.3		X		
465	20.1				X	495	23.7		X			525	30.8		X		
466	17.7				X	496	27.9		X			526	33.7		X		
467	15				X	497	31.9		X			527	35.2		X		
468	12.1				X	498	35.4		X			528	35.2				X
469	9.1				X	499	36.2				X	529	32.5				X
470	6.2				X	500	34.2				X	530	27.9				X
471	3.6				X	501	30.2				X	531	23.2				X
472	1.8				X	502	27.1				X	532	18.5				X
473	0.8				X	503	26.6		X			533	13.8				X
474	0	X				504	28.6		X			534	9.1				X
475	0	X				505	32.6		X			535	4.5				X
476	0	X				506	35.5		X			536	2.3				X
477	0	X				507	36.6				X	537	0	X			
478	0	X				508	34.6				X	538	0	X			
479	0	X				509	30				X	539	0	X			
480	0	X				510	23.1				X	540	0	X			

WMTC. part 1. applicable for vehicles with a maximum design vehicle speed (where applicable truncated at 45 km/h & 50km/h. respectively), cold or warm, 541 to 600 s.

<i>time in s</i>	<i>roller speed in km/h</i>	<i>phase indicators</i>			
		<i>stop</i>	<i>acc</i>	<i>cruise</i>	<i>dec</i>
541	0	X			
542	2.8		X		
543	8.1		X		
544	14.3		X		
545	19.2		X		
546	23.5		X		
547	27.2		X		
548	30.5		X		
549	33.1		X		
550	35.7		X		
551	38.3		X		
552	41		X		
553	43.6			X	
554	43.7			X	
555	43.8			X	
556	43.9			X	
557	44			X	
558	44.1			X	
559	44.2			X	
560	44.3			X	
561	44.4			X	
562	44.5			X	
563	44.6			X	
564	44.9			X	
565	45			X	
566	45			X	
567	45			X	
568	45			X	
569	45			X	
570	45			X	
571	45			X	
572	45			X	
573	45				

time in s	roller speed in km/h	phase indicators			
		stop	acc	cruise	dec
574	45				
575	45				
576	42.3				X
577	39.5				X
578	36.6				X
579	33.7				X
580	30.1				X
581	26				X
582	21.8				X
583	17.7				X
584	13.5				X
585	9.4				X
586	5.6				X
587	2.1				X
588	0	X			
589	0	X			
590	0	X			
591	0	X			
592	0	X			
593	0	X			
594	0	X			
595	0	X			
596	0	X			
597	0	X			
598	0	X			
599	0	X			
600	0	X			

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ON BEHALF OF
AUTOMOTIVE INDUSTRY STANDARDS COMMITTEE

UNDER
CENTRAL MOTOR VEHICLES RULES - TECHNICAL STANDING COMMITTEE

SET-UP BY
MINISTRY OF ROAD TRANSPORT & HIGHWAYS
(DEPARTMENT OF ROAD TRANSPORT & HIGHWAYS)
GOVERNMENT OF INDIA

9th April 2026

Amendment No 1 (08/2024)
To
AIS-049(Rev.1):2016 Electric Power Train Vehicles - CMVR Type Approval
for Electric Power Train Vehicles

1.0 Page No 3/19

Add new clause 3.4 after clause 3.3:

3.4 Coast Down test for M & N Category Electric Power Train Vehicles

Coast down tests for coefficient of aerodynamic and rolling resistance shall be done as per AIS 137 standard as amended from time to time for M & N Category Electric Power Train Vehicles till the time revision of AIS 039 Rev 1:2015 and AIS 040 (Rev 1):2015 is notified.

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UNDER
CENTRAL MOTOR VEHICLES RULES - TECHNICAL STANDING COMMITTEE

SET-UP BY
MINISTRY OF ROAD TRANSPORT & HIGHWAYS
(DEPARTMENT OF ROAD TRANSPORT & HIGHWAYS)
GOVERNMENT OF INDIA

7th August 2024

AUTOMOTIVE INDUSTRY STANDARDS

**Electric Power Train Vehicles -
CMVR Type Approval for Electric
Power Train Vehicles**

(Revision 1)

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THE AUTOMOTIVE RESEARCH ASSOCIATION OF INDIA
P.B. NO. 832, PUNE 411 004

ON BEHALF OF
AUTOMOTIVE INDUSTRY STANDARDS COMMITTEE

UNDER
CENTRAL MOTOR VEHICLE RULES – TECHNICAL STANDING COMMITTEE

SET-UP BY
MINISTRY OF ROAD TRANSPORT & HIGHWAYS
(DEPARTMENT OF ROAD TRANSPORT & HIGHWAYS)
GOVERNMENT OF INDIA

March 2016

Status chart of the Standard to be used by the Purchaser for updating the record

Sr.	Corrigenda	Amend- ment	Revision	Date	Remark	Misc.

General Remarks:

Introduction

The Government of India felt the need for a permanent agency to expedite the publication of Standards and development of test facilities in parallel when the work of preparation of Standards is going on, as the development of improved safety critical parts can be undertaken only after the publication of the Standard and commissioning of test facilities. To this end, the Ministry of Surface Transport (MOST) has constituted a permanent Automotive Industry Standard Committee (AISC) vide order no. RT-11028/11/97-MVL dated September 15, 1997. The Standards prepared by AISC will be approved by the permanent CMVR Technical Standing Committee (CTSC) after approval, The Automotive Research Association of India, (ARAI), Pune, being the secretariat of the AIS Committee, has published this Standard. For better dissemination of this information, ARAI may publish this document on their website.

This Standard prescribes the applicability of CMVR rules and corresponding Standards for Electric Power Train Vehicles.

The AISC panel and the Automotive Industry Standards Committee (AISC) responsible for preparation of this standard are given in Annex F and Annex G respectively.

**Electric Power Train Vehicles - CMVR Type Approval for
Electric Power Train Vehicles**

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**Electric Power Train Vehicles - CMVR Type Approval for
Electric Power Train Vehicles**

**1.0 APPLICABILITY OF THE RULES PRESCRIBED IN CENTRAL
MOTOR VEHICLES (AMENDMENT) RULES, 1989 (CMVR)**

- 1.1 The following rules are not applicable for Electric Power Train Vehicles

Rule No.	Subject
112	Exhaust gases
113	Location of Exhaust Pipes.
114	Exhaust pipes of public service vehicles.
115, (A), (B), (C), (D)	Emission of smoke, vapour, etc. from motor vehicles.
116	Test for smoke emission level and carbon monoxide
120(1)	Silencers

- 1.2 The following rules are applicable, after taking into consideration the amendments listed in Annex A.

Rule No.	Subject
2	Definitions
96(4)	Brakes
101 (1)	Power operated windscreen Wiper
102	Signaling devices, direction indicators and stop lights.
103	Position of the indicator.
104	Fitment of reflectors.
105	Lamps
109	Parking light
110	Lamps on three-wheelers
120 (2)	Noise Standards
122	Embossment of the Chassis number and Engine number in the case of Electric Power Train Vehicles, motor number and month of manufacture
125(1)	Safety-belt, Collapsible steering column, auto-dipper and padded dashboards

- 1.3 All other rules of CMVR shall be applicable as detailed in each rule.

2.0 APPLICABILITY OF STANDARDS NOTIFIED UNDER RULE 124

2.1 The following standards are not applicable to electric power train vehicles

Standard No.	Subject
IS:12056:1987, AIS-033/2001 and IS 14681: 1999	Fuel Tanks
IS 14283: 1995	The accelerator control system
IS 10791: 1983	Control Cables
IS 13942: 1994	External Projection for two wheeled

2.2 Gradeability test (AIS-003)

This test would be conducted once, for electric powertrain vehicles at 60% state-of-discharge condition of the REESS (Rechargeable Energy Storage System).

The gradeability requirement in case of electric power train vehicles shall be as follows.

Two wheeled electric power train vehicles having net motor power less than or equal to 1000 W: 3.81°

All other electric power train vehicles: 7°

2.2 All other notified standards shall be applicable as per provisions of CMVR 1989.

3.0 STANDARDS REFERRED TO IN VARIOUS RULES OF CMVR

3.1 Brake performance:

The changes needed in the test procedure (IS 14664 and IS 11852 as applicable) for evaluating the prescribed performance requirements of braking systems for Electric Power Train Vehicles, to take into account regenerative braking system, are given in Annex B.

Note: Certain amendments to IS 14464 have been agreed in the AISC for incorporating the requirements of three wheelers with GVW between 1 and 1.5 T. These are given in Annex C, which shall be taken into account for compliance requirements.

3.2 Measurement of pass by noise level:

In IS 3028, the approach speed is specified in terms of the engine rpm in certain cases of vehicles. In case of electric power train vehicle the following are the applicable approach speeds:

3.2.1 50km/h or 75% of the maximum design speed of the vehicle whichever is lower, in the following types of vehicles

- a) Two wheelers with maximum design speed exceeding 45km/hr, with manual shift gearbox (Para 8.2.1 of IS 3028:1998.)
- b) Three wheelers with manual shift gearbox (Para 8.3 a) of IS 3028:1998).
- c) Four wheeled and multi axles vehicles with manual gearbox or an automatic transmission with a manual over ride. (Para 8.4.1 of IS 3028:1998.). Condition given in Para 8.4.2 shall apply.

3.2.2 In all other cases the approach speed shall be same as prescribed in IS 3028:1998.

3.3 EMC Test

The EMC test shall be applicable as per AIS-004 (Part 3) as applicable for the category of the electric power train vehicle on the date of application.

4.0 DEFINITIONS

Definitions for electric power train vehicles are enclosed in Annexure-C.

5.0 TECHNICAL SPECIFICATIONS:

The information to be submitted by the manufacturer for type approval shall contain the information given in the revised version of AIS-007 (Rev.5)

Note: The requirements specific to Electric Power Train Vehicles are given in Table 13 of AIS-007 (Rev.5)

6.0 CHANGES IN THE TECHNICAL SPECIFICATIONS ALREADY TYPE APPROVED

6.1 Every modification pertaining to the information declared in accordance with paragraph 5 shall be intimated by the manufacturer to the testing agency.

6.2 If the changes are in parameters not related to the provisions, no further action need be taken.

If the changes are in parameters related to the provisions, the testing agency shall then consider, whether,

- a) the model with the changed specifications still complies with provisions;
- or,
- b) any further verification is required to establish compliance.

For considering whether any further verification is required or not, guidelines given in Annexure D shall be used for the electrical requirements. For other cases, the guide lines given in the individual standard shall be applicable.

- 6.3 In case of 6.2 b), verification for only those parameters which are affected by the modifications needs to be carried out.
- 6.4 In case of fulfillment of criterion of para 6.2 a) or after results of further verification as per para of 6.2 b) are successful, the approval of compliance shall be extended for the changes carried out.

7.0 TRANSITIONAL PROVISIONS

- 7.1 General guidelines for transitional provisions for this standard shall be as per AIS-000, as amended from time to time, as applicable, with the following requirement.
- 7.2 Type approvals issued for compliance to AIS-049 - 2003 shall be extended to approval of AIS-049 (Rev 1) - 2016 however manufacturer may request a retest.

8.0 VALIDITY OF ANNEXES

It is expected that in due course of time the details given in Annexes A, to D would be incorporated in CMVR and other related documents referred in this chapter. Once such an incorporation takes place, the details given in these annexes would be automatically defunct.

Annex: A
(See para 3.1)

Requirements of Brake and Braking Systems for Electric Power Train Vehicles

A1.0 Definitions:

- A1.1 **“Electric regenerative braking system”** means a braking system, which allows the use of the vehicle’s drive motor(s) to convert the kinetic energy of the vehicle into electrical energy during deceleration.
- A1.2 **“Electric regenerative brake control”** means a device which modulates the action of the electric regenerative braking system
- A1.3 **“Electric regenerative braking system of Category A”** means an electric regenerative braking system, which is not part of the service braking system.
- A1.4 **“Electric regenerative braking system of Category B”** means an electric regenerative braking system, which is part of the service braking system.

A2.0 General requirements of electric regenerative braking systems:

A electric power train vehicle, if fitted with an electric regenerative braking system, shall comply with the following requirements:

- A2.1 Vehicles fitted with Electric regenerative braking system of Category A:

The electric regenerative braking shall be only activated by accelerator control and/or the gear neutral position.
- A2.2 Vehicles fitted with Electric regenerative braking system of Category B:

A2.2.1 It shall not be possible to disconnect partially or totally one part of the service braking system other than by an automatic device
- A2.2.2 The service braking system control shall also actuate the action of the electric regenerative braking system simultaneously.
- A2.2.3 The service braking system shall not be adversely affected by the disengagement of the motor(s) or gear ratio used, except during the short duration of operation of gear shifting.

A3.0 Requirements of two wheelers:

- A3.1 The requirements covered in IS 14664:1999 shall be complied with.
- A3.2 If the brake control (front or rear or both) actuates the electric regenerative brake system (Category B), the prescribed performance requirements shall be complied with the use of the electric regenerative system.
- The performance requirements may be verified without the use of the electric regenerative system if so desired by the manufacturer.
- A3.3 If the brake control (front or rear or both) does not actuate the electric regenerative brake system (Category A) the prescribed performance requirements shall be obtained without the use of the electric regenerative system.

A4.0 Requirements of three wheelers:

- A4.1 The requirements covered in IS 14664:1999 shall be complied with.
- A4.2 In the case of three wheelers with GVW not exceeding 1 ton (Category 1 & 2 as defined in IS 14664:1999) the conditions of regenerative braking system shall be same as those given in para A3.2 and A3.3
- A4.3 For three wheelers with GVW exceeding 1 ton (Category 3 as defined in IS 14664:1999) the conditions of regenerative braking system shall be same as those given in para A5.2. and A5.3.
- A4.4 For the purpose of classification only, GVW shall be considered ignoring the weight of the traction batteries. The tests shall be conducted with the full GVW, i.e. including the weight of the traction batteries also.
- A5.0 Requirements of vehicles with more than three wheels:
- A5.1 The requirements given in the applicable parts of IS:11852 shall be complied with.
- A5.2 If the service brake control actuates the electric regenerative brake system (Category B), the prescribed performance requirements shall be obtained with the electric regenerative system.
- A5.2.1 The specified secondary brake prescribed performance requirements shall be met with the failure of the regenerative braking simulated without any other failure in the service brake system.
- A5.3 If the service brake control does not actuate the electric regenerative brake system (Category A) the prescribed performance requirements shall be obtained without the use of the electric regenerative system.

Annex: B

Proposed Amendments to IS 14464: 1999 for Braking System to incorporate the Features for 3 Wheelers of 1 to 1.5T GVW

- B1 Para 1.1 Page 1:
Substitute “of all types of two and three wheeled motor vehicles” for the words “of all types of two wheeled motor vehicles and three wheeled motor vehicles whose maximum gross vehicle weight does not exceed 1500kg”
- B2 Para 4.2 Page 2:
a) In para b) add the following words at the end: “and the gross vehicle weight not exceeding 1 tonne.”
b) Add the following: “c) Category 3--- Three wheeled motor vehicles with gross vehicle weight exceeding 1.0 T.”
- B3 Para 7.11, page 3:
Substitute “Every three wheeled vehicle of Category 1 and 2 shall be fitted with a braking system complying with the requirements given in para a) or b) below. Every three wheeled vehicle of category 3 shall be fitted with a braking system complying with the requirements of para b) below.” for the words “Every three wheeled vehicle shall be fitted with either.”
- B4 Para 7.11. sub-para b) Page 3 :
Add the following at the end of sub-para 5)
“6) In case of three wheelers of category 3, if the requirement given in 7.11(b)(1) above is met by using the same control, the parking brake device shall be so designed that it can be actuated by the driver when the vehicle is in motion with at least one hand on the steering control and shall comply with the requirements for P type dynamic parking brake test given in Table 2. The force exerted on the braking control shall not exceed the values specified in 9.2. Compliance with this requirement shall be deemed to have met if the braking performance has been achieved once.”
- B5 Para 8.1.1 Page 4:
Add the following at the end: “In case of 3 wheelers of Category 3, the P type dynamic parking brake test as specified in 7.11.b) 6 shall also be carried out.”
- B6 Para 9.2 Page 4:
Replace the heading on the second and third columns by “3 wheelers of category 1 & 2” and “3 wheelers of category 3”, respectively.

B7 Table 2 Page 5:

a) Substitute the words “*For Category 2 & 3*” for the words “For category 2” appearing in the first row.

b) Add the following row after entry no. vi)

“vii	Laden	P type dynamic parking brake test {see 7.11.b) 6) for category 3	--	--	--	30	0.1V +V ² / 40	1.5
------	-------	--	----	----	----	----	---------------------------------	-----

B8 Para 12.14 Page 9:

Add the following as a new para: “12.14.3 In the case of P type Dynamic parking brake test {see 7.11.b) 6) for category 3, one test is sufficient”

Annex: C Definitions

- C1.0 This Chapter Annexure gives the definitions of the terminology used for Electric Power Train Vehicles.
- C1.1 **“Access Probe”** means a test probe simulating in a conventional manner a part of a person or a tool, or the like, held by a person to verify adequate clearance from hazardous part.
- C1.2 **"Active driving possible mode"** means the vehicle mode when application of pressure to the accelerator pedal (or activation of an equivalent control) or release of the brake system will cause the electric power train to move the vehicle.
- C1.3 **“Approval of a type of Electric power train vehicle”** means the approval of a type of electric power train vehicle regarding construction and functional safety requirements specific to the use of electric energy.
- C1.4 **“Auxiliary battery”** means the battery unit whose reserve of energy is used only for the auxiliary network supply.
- C1.5 **“Auxiliary network”** means the assembly of auxiliary electric equipment with similar functions to the one used on vehicles equipped with an internal combustion engine.
- C1.6 **“Barrier”** means the part providing protection against direct contact to the live parts from any direction of access.
- C1.7 **"Cell"** means a single encased electrochemical unit containing one positive and one negative electrode which exhibits a voltage differential across its two terminals.
- C1.8 **"Conductive connection"** means the connection using connectors to an external power supply when the rechargeable energy storage system (REESS) is charged.
- C1.9 **"Coupling system for charging the Rechargeable Energy Storage System (REESS)"** means the electrical circuit used for charging the REESS from an external electric power supply including the vehicle inlet.
- C1.10 **“Coupling system”** means all the parts used to connect the vehicle to an external electric power supply (alternative or direct current supply).
- C1.11 **"Chassis connected to the electric circuit"** means AC and DC electric circuits galvanically connected to the electrical chassis."
- C1.12 **“Direct contact”** means contact of persons with the live parts.

- C1.13 **“Drive direction control unit”** means a specific device physically actuated by the driver in order to select the drive direction (forward or backward), in which the vehicle will travel if the accelerator is actuated.
- C1.14 **“Drive Train”** means specific components of power train, such as the traction motors, electronic control of the traction motor, the associated wiring harness and connectors.
- C1.15 **“Degree of protection”** means the extent of protection provided by an enclosure against access to hazardous parts against ingress of solid foreign objects and/or against ingress of water and verified by standardized test methods.
- C1.16 **“Electrical chassis”** means a set made of conductive parts electrically linked together, whose potential is taken as reference.
- C1.17 **“Electrical circuit”** means an assembly of connected live parts which is designed to be electrically energized in normal operation.
- C1.18 **“Electronic converter”** means a device capable of controlling and/or converting electric power for electric propulsion.
- C1.19 **“Electric energy conversion system”** means a system that generates and provides electric energy for electric propulsion.
- C1.20 **“Enclosure”** means the part enclosing the internal units and providing protection against direct contact from any direction of access.
- C1.21 **“Exposed conductive part”** means the conductive part which can be touched under the provisions of the protection IPXXB, and which becomes electrically energized under isolation failure conditions. This includes parts under a cover that can be removed without using tools.
- C1.22 **“Electric power train”** means a system consisting of one or more electric energy storage devices (e.g. a battery, electromechanical flywheel or super capacitor), one or more electric power conditioning devices and one or more electric machines that convert stored electric energy to mechanical energy delivered at the wheels for propulsion of the vehicle;
- C1.23 **“External electric power supply”** means an alternating current (AC) or direct current (DC) electric power supply outside of the vehicle.
- C1.24 **“High Voltage”** means the classification of an electric component or circuit, if its working voltage is $> 60 \text{ V}$ and $\leq 1500 \text{ V DC}$ or $> 30 \text{ V}$ and $\leq 1000 \text{ V AC}$ root mean square (rms).

- C1.25 **"High voltage bus"** means the electrical circuit, including the coupling system for charging the REESS that operates on high voltage.
- Where electrical circuits, that are galvanically connected to each other, are galvanically connected to the electrical chassis and the maximum voltage between any live part and the electrical chassis or any exposed conductive part is ≤ 30 V AC and ≤ 60 V DC, only the components or parts of the electric circuit that operate on high voltage are classified as a high voltage bus."
- C1.26 **"Live parts"** means the conductive part(s) intended to be electrically energized in normal use.
- C1.27 **"On-board isolation resistance monitoring system"** means the device which monitors the isolation resistance between the high voltage buses and the electrical chassis.
- C1.28 **"Open type traction battery"** means a liquid type battery requiring refilling with water and generating hydrogen gas released to the atmosphere.
- C1.29 **"Rechargeable Energy Storage System (REESS)"** means the rechargeable energy storage system that provides electric energy for electric propulsion. The REESS may include subsystem(s) together with the necessary ancillary systems for physical support, thermal management, electronic control and enclosures.
- C1.30 **"Subsystem"** means any functional assembly of REESS components.
- C1.31 **"Type of REESS"** means systems which do not differ significantly in such essential aspects as:
- (a) The manufacturer's trade name or mark;
 - (b) The chemistry, capacity and physical dimensions of its cells;
 - (c) The number of cells, the mode of connection of the cells and the physical support of the cells;
 - (d) The construction, materials and physical dimensions of the casing and
 - (e) The necessary ancillary devices for physical support, thermal management and electronic control.
- C1.32 **"Service disconnect"** means the device for deactivation of the electrical circuit when conducting checks and services of the REESS, fuel cell stack, etc.

- C1.33 **"State of Charge (SOC)"** means the available electrical charge in a tested-device expressed as a percentage of its rated capacity.
- C1.34 **"Solid insulator"** means the insulating coating of wiring harnesses provided in order to cover and protect the live parts against direct contact from any direction of access; covers for insulating the live parts of connectors, and varnish or paint for the purpose of insulation.
- C1.35 **"Net power"** means the power obtained on a test bench at the end of the crankshaft or its equivalent at the corresponding motor speed with the auxiliaries listed in Table 1 of AIS-041 (Rev.1) and determined under reference atmospheric condition;
- C1.36 **"Maximum 30 minutes power"** means the maximum net power of an electric drive train at DC voltage , which a drive train can deliver over a period of 30 minutes as an average.
- C1.37 **"Standard-production equipment"** means equipment provided by the manufacturer for a particular application.
- C1.38 **"Electric range"**, for vehicles powered by an electric power train only, means distance that can be driven electrically on one fully charged REESS.
- C1.39 **"Hazardous Live Part"** means a live part, which, under certain conditions of external influences can give an electric shock (see IEC 536, at present Document 64 (CO) 196).
- C1.40 **"Hazardous Mechanical Part"** means a moving part, other than a smooth rotating shaft, that is hazardous to touch.
- C1.41 **"Hazardous part"** means a part that is hazardous to approach or touch.
- C1.42 **"Indirect contact"** means contact of persons or livestock with exposed conductive parts.
- C1.43 **"IP code"** means a coding system to indicate the degrees of protection provided by an enclosure against access to hazardous parts, ingress of solid foreign objects, ingress of water to give additional information in connection with such protection.
- C1.44 **"Live parts"** means conductive part(s) intended to be electrically energized in normal use.
- C1.45 **"Luggage compartment"** means the space in the vehicle for luggage accommodation, bounded by the roof, hood, floor, side walls, as well as by the barrier and enclosure provided for protecting the power train from direct contact with live parts, being separated from the passenger compartment by the front bulkhead or the rear bulk head.

- C1.46 **“Maximum Mass”** means the technically permissible maximum mass declared by the Manufacturer. Also referred as GVW.
- C1.47 **“Maximum Thirty Minute Power”** means the maximum net power at wheels of an electric vehicle drive train at appropriate rated voltage, which the vehicle drive train can deliver over a period of 30 minutes as an average.
- C1.48 **“Maximum Net Power”** means the power obtained at the wheels of electric vehicle when tested on chassis dynamometer or at Motor shaft when measured at bench dynamometer at corresponding vehicle/motor speed at reference atmospheric conditions and full load on wheels of vehicle/motor.
- C1.49 **“Nominal voltage”** means the root –mean – square (r.m.s.) value of the voltage specified by the Manufacturer, for which the electrical circuit is designed and to which its characteristics are referred.
- C1.50 **“Object Probe”** means a test probe simulating a solid foreign object to verify the possibility of ingress into an enclosure.
- C1.51 **“Off-board charger”** means an energy electronic converter used for charging REESS from an external power supply (mains network) and which is not the integral part of the vehicle.
- C1.52 **“On-board charger”** means an energy electronic converter linked by construction to the vehicle and used for charging the traction REESS from an external electric power supply (mains network).
- C1.53 **“Opening”** means a gap or aperture in an enclosure, which exists or may be formed by the application of a test probe at the specified force.
- C1.54 **“Passenger compartment”** means the space for occupant accommodation, bounded by the roof, floor, side walls, doors, window glass, front bulkhead and rear bulkhead, or rear gate, as well as by the barriers and enclosures provided for protecting the power train from direct contact with live parts.
- C1.55 **“Protection degree”** means the protection provided by a barrier/enclosure related to the contact with live parts by a test probe, such as a test finger (IPXXB) or a test wire (IPXXD).
- C1.56 **“Power train”** means the system of energy storage device(s), energy converter(s) and transmission(s) that converts stored energy to mechanical energy delivered at the wheels for propulsion of the vehicle;
- C1.57 **“Pure electric vehicle”** means vehicle powered by an electric power train only;

C1.58 **“Reference Mass”** means

- i) Two wheeled vehicles: unladen mass increased by a figure of 75 kg
- ii) Three wheeled vehicles, L7 and M1, M2 (with GVW upto 3500 kg) category vehicles: unladen mass increased by a figure of 150 kg
- iii) N category vehicles: unladen mass increased by a figure of 50% of the pay load”.
- iv) For M2 (with GVW above 3500 kg) and M3 vehicles: unladen mass increased by 75 kg x 50% of Seating Capacity.

C1.59 **“Unladen Mass”** means the mass of the vehicle in running order without driver, crew, passengers or load, but with the fuel tank full (if any), cooling liquid, service and traction batteries, oils, onboard charger, portable charger, tools and spare wheel, whatever is appropriate for the vehicle considered and if provided by the manufacturer of the vehicle.

C1.60 **“Vehicle type”** means Electric Power Train vehicles, which do not differ in such essential aspects as:

dimensions, structure, shape and nature of constituting materials; installation of the power system components , REESS or REESS packs, tyres, unladen mass; nature and type of electric and electronic components.

Also **"Vehicle type"** means vehicles which do not differ in such essential aspects as:

- (a) Installation of the electric power train and the galvanically connected high voltage bus;
- (b) Nature and type of electric power train and the galvanically connected high voltage components.

C1.61 **"Working voltage"** means the highest value of an electrical circuit voltage root-mean-square (rms), specified by the manufacturer, which may occur between any conductive parts in open circuit conditions or under normal operating condition. If the electrical circuit is divided by galvanic isolation, the working voltage is defined for each divided circuit, respectively.

Annex: D
(See 6.0)
Criteria for Extension

D1.0 In case of following changes, the verification tests which are necessary for establishing compliance are listed below

	Changes of Parameter	Tests to be conducted
D1.1	Type of motor, controller, REESS	Range test, Energy consumption test, Power test, All tests/verifications as per AIS-038 (Rev.1):2015, Gradeability, EMC, and if fitted with regenerative braking of category B, brake test.
D1.2	Layout of REESS, controller, motor	All applicable tests/verifications as per AIS-038 (Rev.1):2015.
D1.3	Working Voltage	All tests
D1.4	Make, model of on-board charger	Energy consumption test as per AIS-039 (Rev.1):2015, Range Test as per AIS-040 (Rev.1):2015
D1.5	Vehicle body construction affecting aerodynamic resistance	Coast-down, range, energy consumption, EMC
D1.6	Wiring harness of Traction REESS	Compliance to 3.1.1.5.3 of AIS-038 (Rev.1):2015
D1.7	In the case of tests affecting range, , energy consumption because of change in reference mass, coast-down parameters etc. the conditions given in MoRTH /CMVR/TAP-115/116 shall be applicable	
D1.8	Changes in parameter affecting general performance requirements of CMVR (E.g. brakes, noise etc.) shall be as per details given in the individual standards.	
D1.9	For changes other than the above, the provisions given in the AIS-017 (Procedure for Type Approval and Certification of Vehicles for Compliance to Central Motor Vehicles Rules) may be followed.	

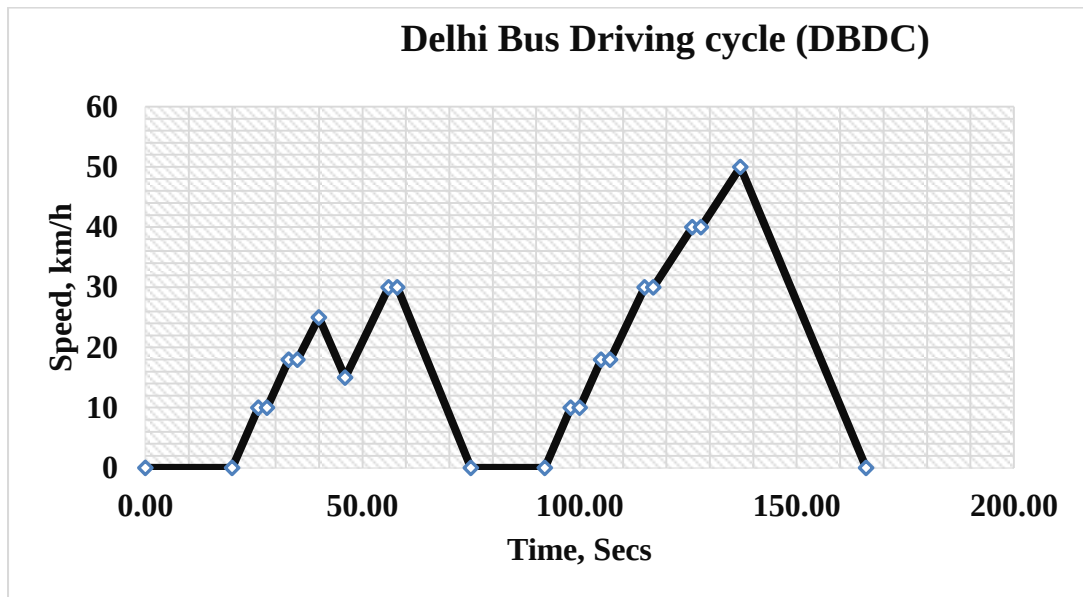
ANNEX E
(See clause 3.4 b of AIS 039 (Rev 1):2015)
DELHI DRIVING CYCLE FOR BUSES

Table 1: Phase wise operation and duration of the Delhi bus Driving cycle (DBDC)

No. of Operation	Operation	Phase	Speed Km/h	Acceleration m/s ²	Duration of Operation , s	Cumulative time, s	Distance, m
1	Idling	1	0	0	20	20	0
2	Acceleration	2	0-10	0.46	6	26	10
3	Cruising	3	10	0	2	28	6
4	Acceleration	4	10-18	0.44	5	33	23
5	Cruising	5	18	0	2	35	10
6	Acceleration	6	18-25	0.39	5	40	31
7	Deceleration	7	25-15	-0.46	6	46	35
8	Acceleration	8	15-30	0.42	10	56	69
9	Cruising	9	30	0	2	58	17
10	Deceleration	10	30-0	-0.49	17	75	79
11	Idling	11	0	0	17	92	0
12	Acceleration	12	0-10	0.46	6	98	10
13	Cruising	13	10	0	2	100	6
14	Acceleration	14	10-18	0.44	5	105	23
15	Cruising	15	18	0	2	107	10
16	Acceleration	16	18-30	0.42	8	115	58
17	Cruising	17	30	0	2	117	17
18	Acceleration	18	30-40	0.31	9	126	71
19	Cruising	19	40	0	2	128	22
20	Acceleration	20	40-50	0.31	9	137	91
21	Deceleration	21	50-0	-0.48	29	166	219
							837

Total Distance = 837 m

Total time = 166 s



ANNEX F
(See introduction)

COMPOSITION OF AISC PANEL ON
Electric Power Train Vehicles - CMVR Type Approval for Electric Power
Train Vehicles

Convener	
Mr. A.A. Deshpande	The Automotive Research Association of India (ARAI)
Members	Representing
Mr. M.M.Desai	The Automotive Research Association of India (ARAI)
Mr. D. P. Saste/ Mr. Karthikeyan K (Alternate)	Central Institute of Road Transport (CIRT)
Representative from	International Centre for Automotive Technology (ICAT)
Mr. Vinod Kumar	Vehicle Research & Dev. Estt. (VRDE)
Dr. N. Karuppaiah	National Automotive Testing and R&D Infrastructure Project (NATRIP)
Mr. K. K. Gandhi / Shri Sourabh Rohila	Society of Indian Automobile Manufacturers (SIAM)
Shri. S Ravishankar	Ashok Leyland Ltd. – Technical Center (SIAM)
Mr. T. M. Balaraman/Mr. Adish Agrawal	Hero MotoCorp Ltd.(SIAM)
Mr. Rajendra Khile	General Motors Technical Center India Pvt. Ltd. (SIAM)
Mr. Kiran Mulki	Mahindra & Mahindra Ltd. (SIAM)
Mr. K. Kiran Kumar	Mahindra Reva Electric Vehicles Pvt. Ltd. (SIAM)
Mr. Vijeth R Gatty	Toyota Kirloskar Motor Pvt. Ltd. (SIAM)
Mr. Firoz Khan	Tata Motors Ltd (SIAM)
Mr. Vivekraj S	Renault Nissan Pvt Ltd(SIAM)
Mr. Uday Harite	Automotive Components Manufacturers Association of India (ACMA)

ANNEX G
(See Introduction)
COMMITTEE COMPOSITION *
Automotive Industry Standards Committee

Chairman	
Mrs. Rashmi Urdhwareshe	Director The Automotive Research Association of India, Pune
Members	Representing
Representative from	Ministry of Road Transport and Highways, New Delhi
Representative from	Ministry of Heavy Industries and Public Enterprises (Department of Heavy Industry), New Delhi
Shri S. M. Ahuja	Office of the Development Commissioner, MSME, Ministry of Micro, Small and Medium Enterprises, New Delhi
Representative from	National Automotive Testing and R&D Infrastructure Project (NATRiP)
Shri N.K Sharma	Bureau of Indian Standards (BIS), New Delhi
Director	Central Institute of Road Transport(CIRT), Pune
Director	Indian Institute of Petroleum(IIP), Dehra Dun
Director	International Centre for Automotive Technology (ICAT), Manesar, Delhi.
Director	Vehicles Research and Development Establishment (VRDE), Ahmednagar
Shri Shrikant R. Marathe	Former Chairman, Automotive Industry Standards Committee
Representatives from	Society of Indian Automobile Manufacturers (SIAM), New Delhi
Shri T.R.Kesavan	Tractor Manufacturers Association (TMA), New Delhi
Shri Uday Harite	Automotive Components Manufacturers Association of India (ACMA), New Delhi

A. S. Bhale
Member Secretary

The Automotive Research Association of India, Pune

* At the time of approval of this Automotive Industry Standard (AIS)